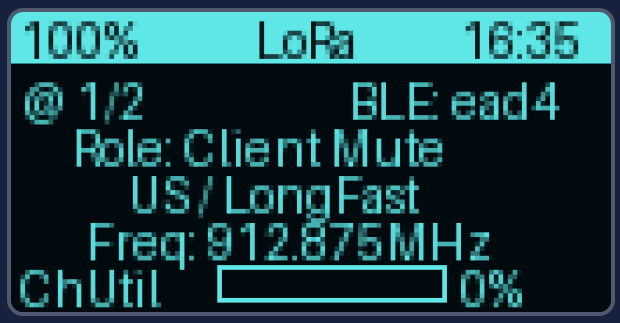
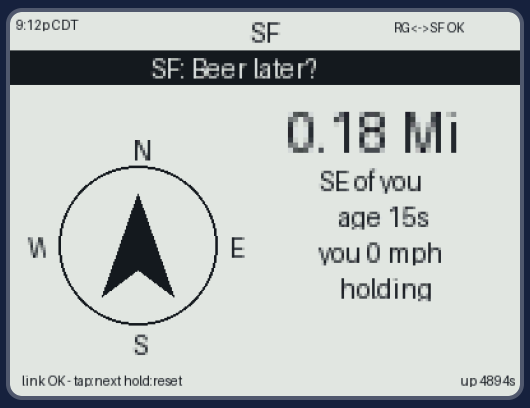
OFF-GRID VEHICLE TRACKER

Device User Guide

Every screen, every button, from power on to power off





This guide walks you from switching a radio on, through every screen and menu you will meet, to switching it off again, with the dashboard display covered along the way. Every screen shown is an exact recreation of what the real device shows, redrawn pixel for pixel so it stays sharp at any size. If a screen is not in this guide, you do not need it.

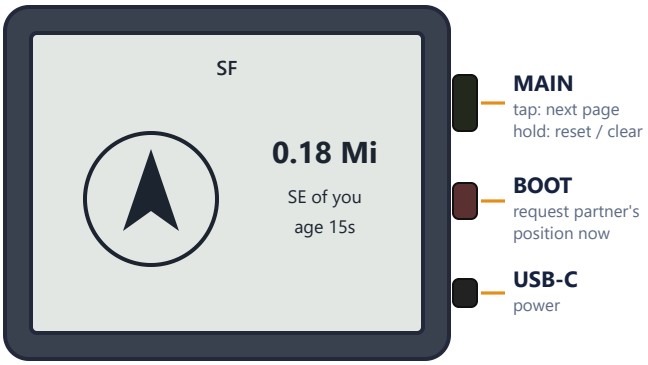
GROUP 9 · CAPSTONE PROJECT LAB · TEXAS TECH UNIVERSITY · SPONSOR: DR. RICHARD GALE · OCTOBER 2026

1 Meet the Devices and Their Buttons

Each vehicle carries two devices. The **radio** is the small green box with the antenna; it does all the talking over the air. The **display** is the larger dashboard screen; it shows you where your partner is. Between them there are exactly four controls, drawn below.



The radio (Heltec V4 in its case). One button navigates everything: short press moves, long press chooses.

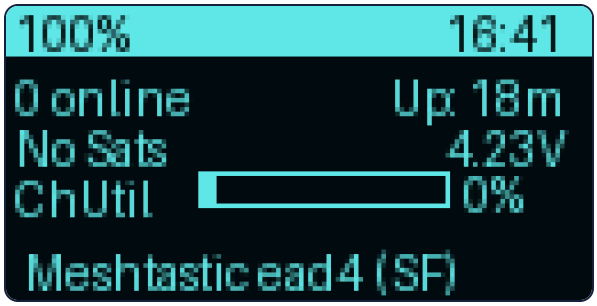


The display (Waveshare reflective panel). Tap to flip pages, hold to reset or clear, BOOT to request the partner's position now.

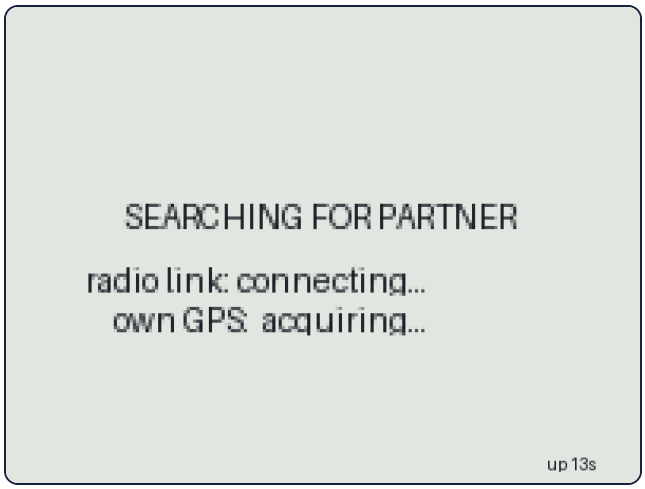
Control	Short press / tap	Long press / hold
Radio, USER button	Next screen; next line inside a menu	Open the action menu for the current screen; confirm the highlighted line
Radio, POWER button		Switch the radio on
Display, MAIN button	Next page (compass, status, messages)	Reset tracking data; on the Messages page, clear the history
Display, BOOT button	Request the partner's position immediately	

> **THE ONE RULE.** Keep both devices charged, and never run the radio's battery completely flat. A fully drained battery can erase the radio's settings, and then the two radios stop understanding each other until a laptop fixes it.

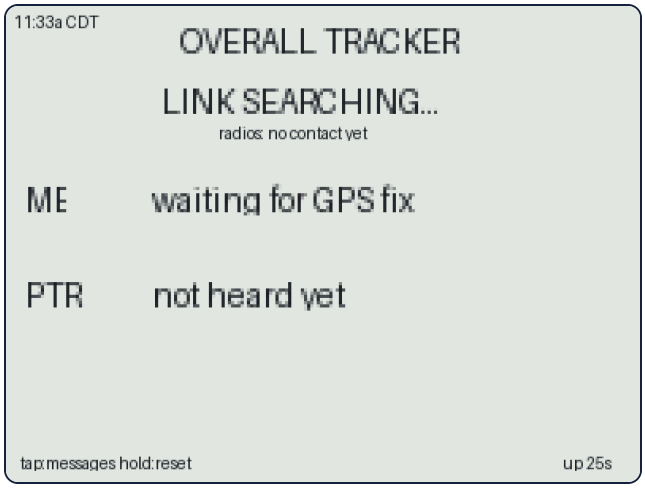
2 Powering On



1 **Hold the radio's power button** until the screen lights, then let it settle for a minute. This home screen is what normal looks like: battery top left, uptime top right, the radio's own name at the bottom, here "Meshtastic ead4 (SF)". Indoors it will say "0 online" and "No Sats"; that is not a fault. Satellites need a view of the sky, and your partner's radio may simply be off.

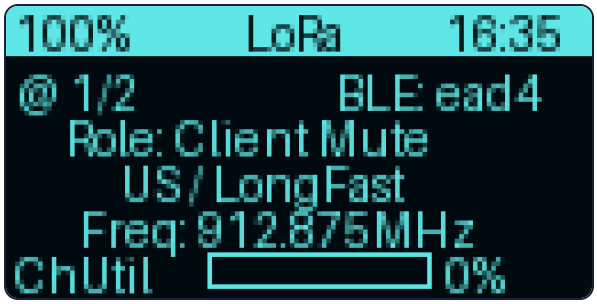


2 **Power the display at the same time**, like turning an ignition key. For its first seconds it shows this searching screen while it connects to its radio over Bluetooth. The radio only accepts the display's connection during its first minute after power on, which is why the two should always be switched on together. If you miss the window, switch both off and on again together.



3 **Watch the page fill in.** "waiting for GPS fix" clears when the satellites arrive, and "PTR not heard yet" clears the first time your partner's radio is heard on the channel. Outdoors with both vehicles on, the whole page is normally live within a minute or two.

3 The Two-Second Pre-Flight Check

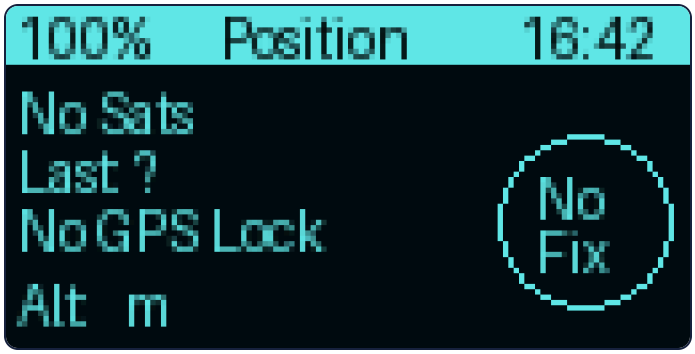


Before every trip, short-press the radio's user button until you reach this **LoRa screen**, and read one number: **Freq: 912.875 MHz**. Check it on both radios. If both say 912.875, the radios can hear each other and everything else will follow. If either shows a different number (906.875 is the usual wrong answer), that radio has lost its settings and needs the recovery procedure in the Technical Manual before the trip.

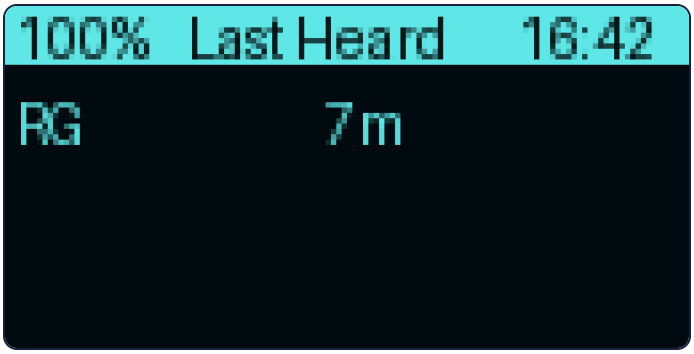
While you are here, the rest of the screen should read Role: Client Mute and US/LongFast, exactly as shown.

4 Reading the Radio's Screens

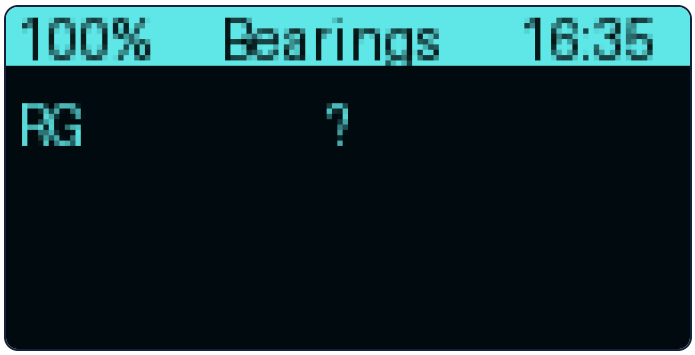
Short presses of the user button walk a loop of status screens. Here is each one and what it tells you.



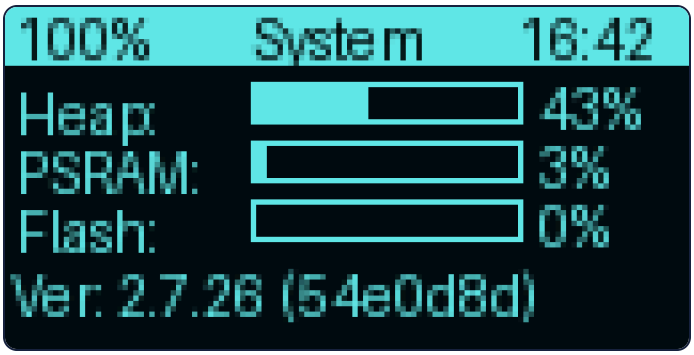
Position. Your own GPS state. "No Sats / No GPS Lock / No Fix" simply means indoors; it fills with coordinates under open sky.



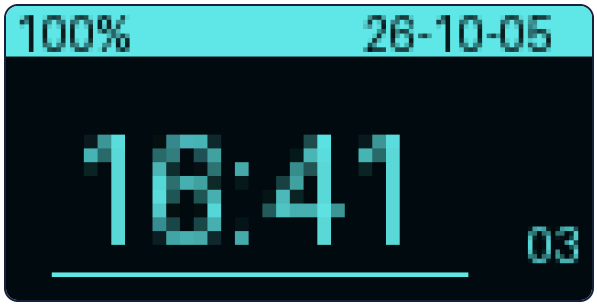
Last Heard. Every radio heard recently and how long ago; here RG was heard 7 minutes ago. The radio's own honesty meter.



Bearings. Direction to other radios once positions are known; a question mark means no position for them yet.



System. Memory and firmware version (2.7.26). Needed only when someone asks what version you run.



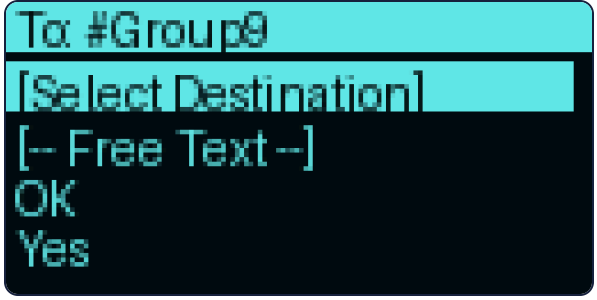
Clock. Time and date arrive from the GPS satellites, so there is nothing to set and the clock is exactly as accurate as the positions are. If the clock is right, the GPS has spoken to this radio.

5 Sending a Message from the Radio

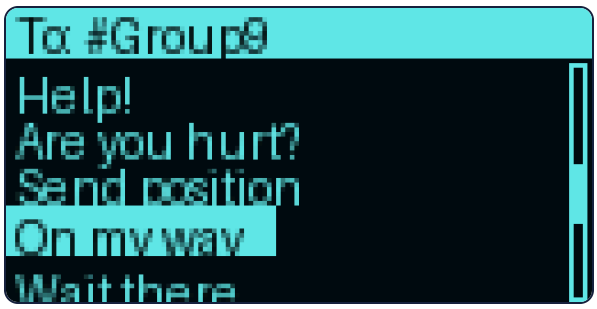
The radio sends ready-made messages so nobody types while driving. The whole flow is five screens, and every press is the same user button: short press to move, long press to choose.



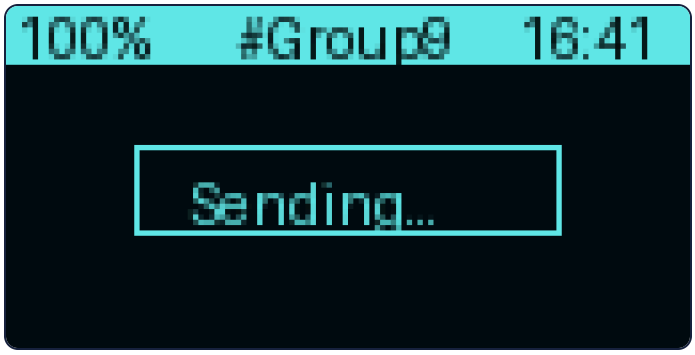
1 From the chat screen, a long press opens the **Message Action** menu. Short-press down to Reply (or View Chats to browse), then long-press to choose.



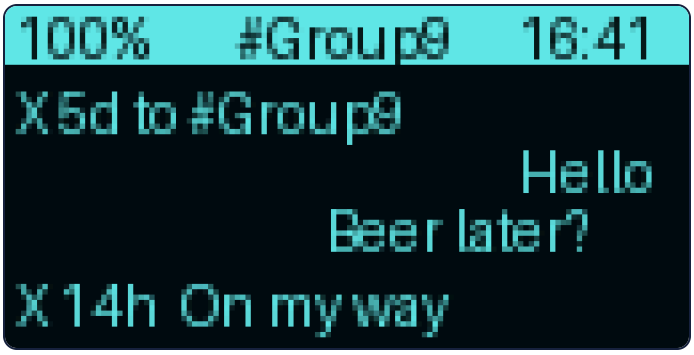
2 Choose the destination. **#Group9** is the shared channel both vehicles hear, and it is the one you want. Skip Free Text; it is for typing with a phone attached, which no one does while driving.



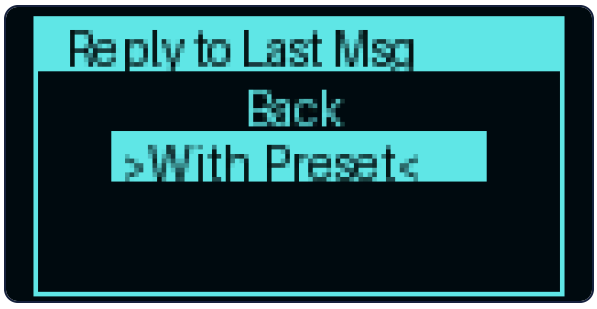
3 Scroll the preset list with short presses. The 21 messages cover the whole conversation: Help!, Are you hurt?, Send position, On my way, Wait there, Turn left, Turn right, Stop, and the rest. Long-press the one you want.



4 **Sending.** The radio shows this for a moment as the message goes out over the air.



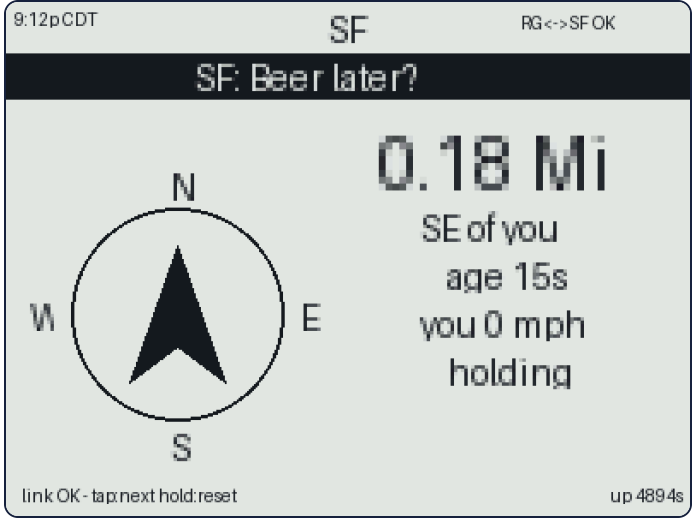
5 **Delivered.** The message joins the thread and pops up on your partner's display within seconds. An X next to a message means no radio has confirmed it yet.



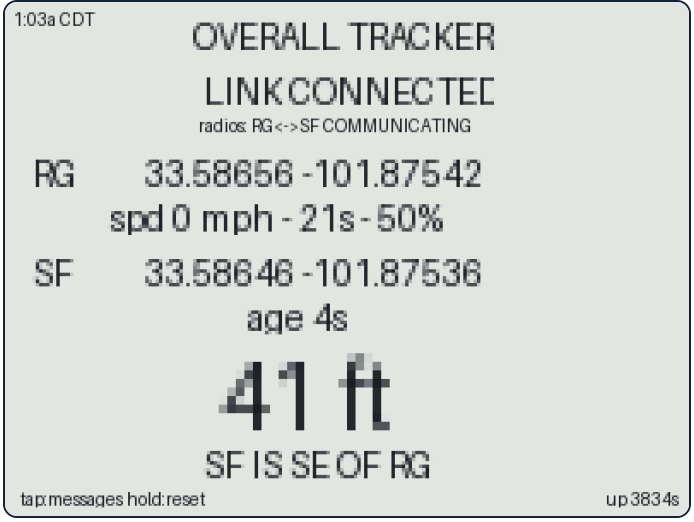
Replying is even shorter. When a message arrives, long-press on the chat screen and choose **Reply to Last Msg, With Preset**. The same preset list opens, aimed back at the sender. Every question in the list has its answer in the list: "Are you hurt?" pairs with "I am good" or "Help!", and "Beer later?" pairs with "Yes".

6 The Display While You Drive

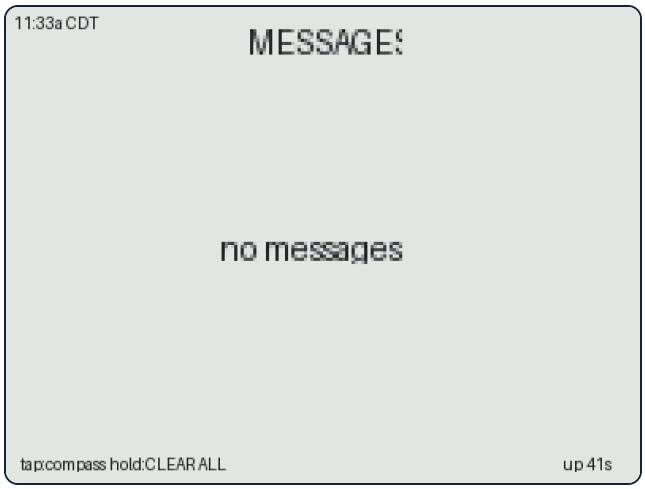
You never need to touch the display while driving; it updates itself twice a second. A tap cycles its three pages.



The compass page. The arrow points at your partner; while driving, straight up means dead ahead. The big number is the distance. An arriving message appears as a dark banner, here "SF: Beer later?", for eight seconds.



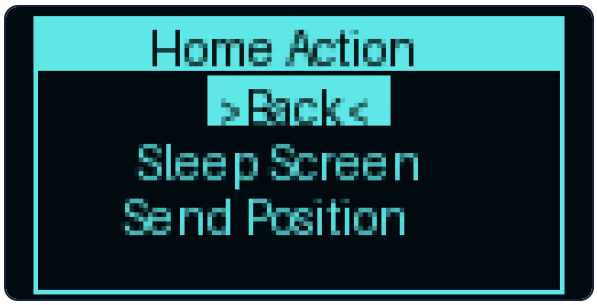
The status page. Both vehicles' coordinates and data ages, the partner radio's battery, the distance in big digits, and the direction in words: "SF IS SE OF RG".



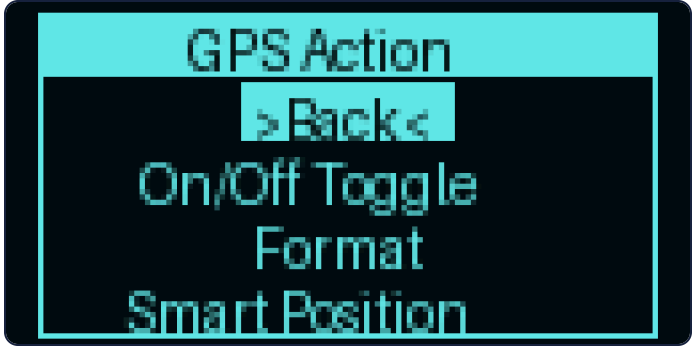
The Messages page keeps the last few received texts, newest first. "no messages" is its honest empty state. Holding the button on this page clears the history; holding it on any other page resets the tracking data, which is harmless because fresh data rebuilds in seconds.

Want your partner's position right now? Press the display's BOOT button. Their radio answers within seconds, and the age number on your screen drops to nearly zero.

7 Menus to Use, Menus to Leave Alone



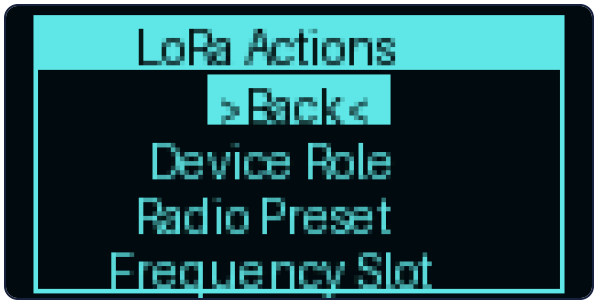
Safe and useful: Home Action. A long press on the home screen offers **Send Position**, which broadcasts your position immediately, and Sleep Screen, which dims the screen to save battery. Both are harmless.



Safe: GPS Action and Clock Action adjust formats only. Leave Smart Position switched on; it is what saves airtime.

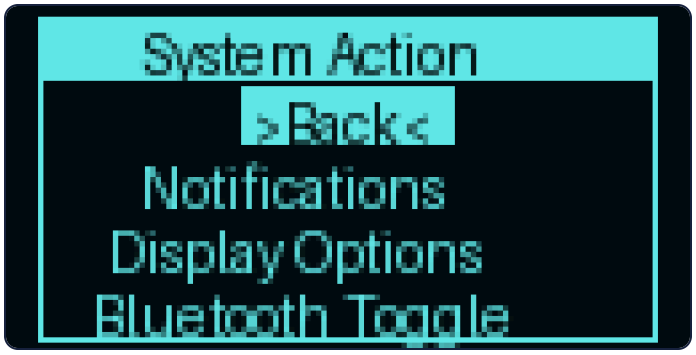


Troubleshooting only: Node Action. Reset NodeDB wipes the list of known radios. Use it only when the Technical Manual says so.

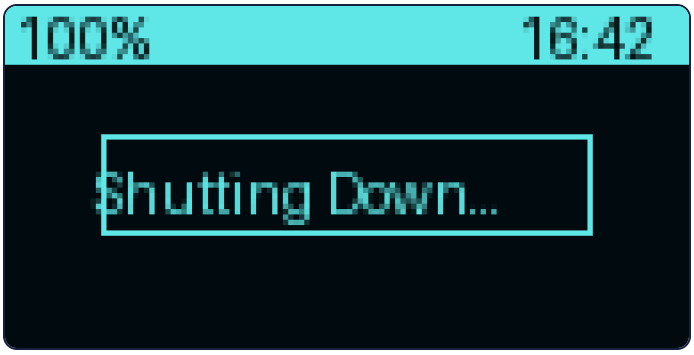


DO NOT CHANGE ANYTHING UNDER LORA ACTIONS. Device Role, Radio Preset, Frequency Slot and LoRa Region are the exact settings that let the two radios hear each other. Changing any of them on one radio makes the pair go silent with no error message anywhere. The same caution applies to Bluetooth Toggle under System Action: switching Bluetooth off strands your display.

8 Powering Off and Charging



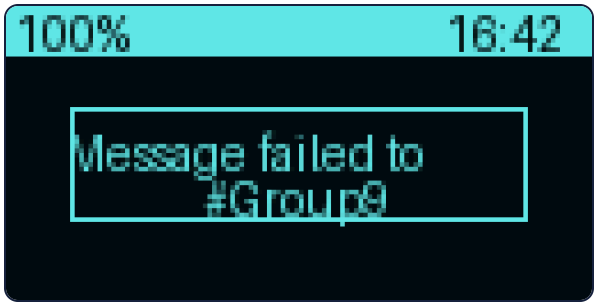
1 Long-press on the home screen and open the **System Action** menu. Short-press down past Notifications and Display Options to Reboot/Shutdown.



2 Choose Shutdown. The radio says "Shutting Down..." and switches off cleanly. A clean shutdown always preserves the settings.

Charge both devices over USB-C after every trip. The radio holds its charge for months when switched off. Power the display from the vehicle or any USB source. Never store the radio with an empty battery.

9 When Something Looks Wrong



"Message failed" means nobody acknowledged the message, usually because the partner's radio is off, out of range, or still booting. The message is not lost forever; send it again once the Last Heard screen shows the partner recently.

What you see	What it means	What to do
0 online, No Sats indoors	Normal. No sky view, partner off.	Nothing. Take it outside and switch the partner on.
Frequency is not 912.875	The radio lost its settings, usually after a dead battery.	Recovery procedure in the Technical Manual (five minutes with a laptop).
Display stuck on "SEARCHING FOR PARTNER"	It missed the radio's connection window.	Switch radio and display off, then on together.
Partner on screen but data age climbing	Distance or terrain is blocking the link.	It heals itself as you get closer. The screen never pretends.
Anything else odd	Electronics being electronics.	Power-cycle both devices: off, count to five, on together.

10 Quick Reference

I want to...	Do this
Start up	Charge check, antennas on, power radio and display together, confirm the display connects.
Pre-flight	Short-press to the LoRa screen on both radios: both must read 912.875.
Send a preset message	Long press on chat, Reply or destination #Group9, pick the preset, long press.
Reply fast	Long press, Reply to Last Msg, With Preset.
Broadcast my position now	Long press on home, Send Position.
Get partner's position now	Press BOOT on the display.
Shut down	Long press on home, System Action, Reboot/Shutdown.
Fix almost anything	Power-cycle both devices together; check 912.875.

Appendix A: Abbreviations and Acronyms

Every short form that appears on the device screens or in this guide, in plain words.

Term	Meaning
BLE	Bluetooth Low Energy. The short wireless link between the display and its own radio inside your vehicle; "BLE: ead4" on the LoRa screen names the radio the display pairs with.
BOOT	The small secondary button on the display. In this system it requests the partner's position immediately.
CDT	Central Daylight Time, the timezone shown on the display's clock corner.
ChUtil	Channel utilization. How busy the shared radio channel has been lately, as a percentage. Single digits is healthy.
Flash / Heap / PSRAM	Three kinds of memory inside the radio, shown on the System screen. You never need to act on them.
GPS	Global Positioning System. The satellites every position fix comes from. Listening to them is free and works anywhere with a sky view.
#Group9	The team's private, encrypted channel. Both radios carry it; messages sent to it reach the partner vehicle.
LoRa	Long Range. The radio technique that carries signals for miles on very little power, at the cost of speaking slowly.
LongFast	The radio speed setting both radios share, the standard compromise between range and speed. Shown as "US/LongFast".
ME / PTR	On the display's status page: your own vehicle (ME) and the partner vehicle (PTR).
MHz	Megahertz, a spot on the radio dial. Both radios must show 912.875 MHz.
NodeDB	The radio's stored list of other radios it has heard. Reset it only during troubleshooting.
OLED	The radio's small glowing screen.
RLCD	Reflective liquid crystal display, the dashboard screen. It reflects daylight instead of glowing, so it is brightest in the sun.
SF and RG	The callsigns of the two radios, written on their cases. Your display shows your partner's callsign.
Client Mute	The radios' configured role: talk and listen normally, never relay. Correct for a two-vehicle system.
USB-C	The oval charging and power plug, the same one most modern phones use.
Ver	Firmware version, shown on the System screen; the radios run Meshtastic 2.7.26.

Off-Grid Vehicle Tracker · Group 9, Texas Tech University ECE Capstone · Companions: User Manual (concepts) and Technical Manual (engineering and recovery procedures). Screen images are exact recreations of the device screens, radio firmware Meshtastic 2.7.26, drawn from photographs of the Group 9 hardware.